

Representation Sensitivity and Cross-Source Generalization in Phishing URL Detection

Final Technical Research Report

Version 1.1.1 · 2026-09-10

Version 1.1.1 adds the previously completed depth-stratified sensitivity analysis as §13.7, with one corresponding threats entry and one Discussion sentence; subsections 13.8-13.13 are renumbered accordingly. Version 1.1 embedded the ten publication figures in the report body and renumbered them by first appearance; all other prose, values, tables and claims are identical to version 1.0. This is the project's authoritative technical research report. It is not the conference or journal manuscript; it is the evidence base from which such a manuscript would be written. Every numeric value below is interpolated at build time from a frozen result file.

1. Executive Summary

What was studied. Whether the performance reported for phishing-URL detectors on a single public benchmark is a property of the detection problem or a property of how the evaluation corpus is represented and composed. The application domain is phishing URL detection; the object of study is evaluation methodology.

Why. Published detectors routinely report near-ceiling accuracy on single benchmarks. If those numbers move substantially when the URL representation or the source composition changes, then a single benchmark score is not a reliable estimate of transferable detection ability, and reporting practice needs to change.

What was found. On the PhiUSIIL benchmark, raw-URL ROC-AUC is 0.9988. Progressive removal of URL surface form reduces this to 0.7297 under full normalisation, and the same pattern replicates across three model families. Decomposition of the observed reduction attributes most of it to residual host and subdomain structure in the PhiUSIIL-phishing cells (share 0.870 in cell A) and substantially less in the PhishTank-phishing cells (0.563 in cell C, with a wide interval). Substituting the phishing corpus while holding the benign population fixed moves normalised performance materially: A-C = -0.1333 (95% CI [-0.1888, -0.0597]) and B-D = -0.1639 ([-0.2599, -0.0734]). The formal four-cell interaction, however, is +0.0413 with an interval including zero, and therefore does not establish differential dependence on the phishing versus benign factor. Masking specific hosting platform identity increased held-out normalised AUC in **60 of 60** seed-cell comparisons. Above-chance normalised discrimination persists when both the phishing and the benign source are replaced ($R^4 \approx 0.6936$ and 0.7061), though that evaluation rests on only 119 independent benign domains. A benchmark-trained raw model flags every one of 11,895 naturally collected legitimate URLs from 362 sites, against a held-out benchmark false-positive rate of 0.00026 (7 / 26,946, $\tau = 0.5$).

What it means. Measured phishing-URL detection performance is substantially representation-sensitive and source-dependent. A benchmark AUC reported at a single representation, on a single source composition, from a single domain-grouped split is not a stable quantity: in this study each of those three choices moved the headline number by more than the margins typically claimed between competing methods.

What remains unresolved. The causal origin of the discrimination that survives full normalisation. It may reflect transferable phishing-related characteristics, source-dependent structure, or a mixture. The experiments constrain this question but do not identify the mechanism, and no result in this report should be read as resolving it.

2. Research Motivation

Phishing-URL detection is commonly evaluated by training and testing on one labelled corpus and reporting accuracy or ROC-AUC. Two properties of that practice are rarely examined. First, a URL carries a great deal of surface form — scheme, `www` prefix, path, query — that is cheap for an adversary to alter and that may differ systematically between how the phishing and legitimate classes of a corpus were assembled. Second, the two classes of a benchmark are typically collected by different processes, so any systematic difference between those processes is available to a learner as discriminative information, independently of whether it characterises phishing.

Neither property is a defect of any particular dataset. Both are structural features of how URL corpora are built, and both bear directly on whether a benchmark score estimates transferable detection ability. This study measures how much of a reported score depends on them.

3. Research Question

To what extent does measured phishing-URL detection performance depend on URL representation and source composition, and what does this imply for the validity of single-dataset benchmark evaluation?

4. Research Objectives

1. Quantify the change in measured performance as URL surface form is removed in a nested sequence, and decompose the observed reduction into interpretable components with uncertainty intervals.
2. Determine whether normalised performance depends on which phishing corpus and which benign population constitute the evaluation set, and separate the two factors in a factorial design.
3. Test whether a model fitted on one phishing corpus generalises to an independently collected one, under a benign population independent of both.
4. Characterise the role of hosting-platform identity in normalised discrimination, distinguishing platform identity from platform membership.
5. Establish the stability of all of the above under repeated domain-grouped resampling, and validate the evaluation pipeline with negative controls.

5. Hypotheses

ID	Hypothesis	Status
H1	Raw benchmark performance on PhiUSIIL is extremely high	Established
H2	Performance is strongly representation-sensitive, degrading substantially under progressive normalisation	Strongly supported
H3	Most of the observed R1→R4 AUC reduction is associated with residual host/subdomain structure	Strongly supported (as a decomposition of the observed reduction)
H4	Normalised discrimination varies across phishing/benign source combinations	Strongly supported
H5	The source effect depends disproportionately on the phishing-source rather than the benign-source factor	Inconclusive — the factorial interaction interval includes zero

ID	Hypothesis	Status
H6	Normalised discrimination persists under independent-benign evaluation	Supported with limitations — 119 benign hold-out domains
HP	Specific hosting-platform identity affects held-out generalisation	Strongly supported — 60/60 comparisons; masking <i>increased</i> AUC

H3 is a statement about how the observed AUC reduction decomposes. It is not a statement that any share of model performance is an artifact. HP is directionally opposite to the pre-registered expectation: masking platform identity improved rather than degraded held-out performance.

6. Scientific Position and Claim Boundaries

The study demonstrates that measured phishing-URL detection performance is substantially representation- and source-dependent. It does **not** establish the causal origin of the discrimination that survives full normalisation.

Four categories are kept distinct throughout this report and are never merged.

A. Direct experimental findings. Quantities produced by an explicit manipulation under a fixed protocol: the representation ablation, the source substitution, the platform-identity masking, the shuffled-label controls.

B. Associational interpretations. Patterns the results support but do not uniquely identify: for example that residual host structure carries most of the observed reduction in the PhiUSIIL-phishing cells.

C. Causal conclusions. Restricted to the manipulation itself. "Masking specific platform identity increased held-out AUC" is causal with respect to masking. It is not a causal claim about why.

D. Unresolved alternatives. Questions the design cannot separate, chiefly the origin of the residual normalised signal, which may reflect transferable phishing-related characteristics, source-dependent structure, or a mixture.

7. Dataset and Source Inventory

See **T01_dataset_inventory** for the full inventory with collection mechanism and per-source caveats.

Source	Role	URLs	U1 domains	U2 domains
PhiUSIIL phishing	phishing class	100,945	43,512	66,278
PhiUSIIL legitimate	benign class	134,850	132,117	132,186
PhishTank (online-valid)	phishing class	74,945	14,321	—
Crawler benign, overlap pool	benign class	21,991	945	945
Crawler benign, disjoint pool	benign class / independent benign	12,427	596	597
Natural legitimate harvest	out-of-distribution stress test	11,895	—	362 sites

The PhiUSIIL benchmark contains 235,795 URLs at 42.8% phishing over 175,509 registrable domains. Its authoritative label convention is `1 = legitimate`, `0 = phishing`; this project recodes to `1 = phishing` throughout, and the orientation was verified empirically before any modelling.

Two properties of the benchmark are load-bearing for the analysis. The PhiUSIIL legitimate class consists entirely of bare `https://www.<domain>` homepages, so the benign class carries a systematic form signature. And the raw feature file contains label-derived columns — `URLSimilarityIndex == 100` identifies the legitimate class almost perfectly — which are excluded: this study uses URL-string features only.

Provenance limitation. The original collection mechanism of the PhiUSIIL legitimate class is not documented at a level this study can verify. No claim is made about how that class was assembled.

The crawler pools differ markedly in reachability. The overlap pool succeeded on 946 of 1,200 queried domains; the disjoint pool on 605 of 1,200. The disjoint pool is also lower-ranked by popularity. Both facts are characterised, not corrected.

8. Data Construction and Filtering

Canonicalisation is applied identically to every source before any comparison. Deduplication is by canonical URL within a source. Deduplication by U2 is applied asymmetrically inside each factorial cell: it is applied to a class only when that class's URLs-per-U2 ratio is at or below 10. Symmetric U2 deduplication was tested and rejected because it collapses the benign class far enough to make low-FPR operating points undefined.

A U1-level deduplication variant retains one URL per public-suffix-only registrable domain, reducing the benchmark from 235,795 to 175,509 URLs (43,467 phishing, 132,042 legitimate; phishing rate 0.2477, retaining 0.7443 of rows). This variant is used for the cross-model replication (§13.4) and for concentration analysis; the factorial design does not use it.

Cross-source evaluation additionally removes the 593 U1 domains that occur in both phishing corpora — predominantly hosting platforms and URL shorteners, carrying 34.28% of PhiUSIIL phishing rows and 59.51% of PhishTank rows — from both corpora in both roles, so that domain-level disjointness holds by construction rather than by argument.

9. Unit Definitions and Leakage Controls

Three registrable-unit definitions are used.

- **U1** — registrable domain under the public-suffix-only definition. This is the grouping unit for every train/test split and for the cluster bootstrap.
- **U2** — registrable domain under the PSL private-aware definition, so that `abc.web.app` and `xyz.web.app` are distinct units.
- **U3** — recognised hosting-platform domains grouped by service identity; unrecognised domains form their own group.

Because U1 is public-suffix-only, all `*.web.app` URLs already share the single group `web.app`. Subdomain-level platform sharing across a split is therefore prevented by the primary protocol, before U3 is considered.

Leakage controls enforced as hard failures at run time: zero U1 overlap between train and test; zero canonical-URL overlap across a split; both classes present in train and in test; for cross-source work, zero U1 and zero canonical overlap between the training corpus and the external test corpus; and, for the character n-gram pipeline, the fitted TF-IDF vocabulary hashed after fitting and re-checked after every external prediction so that target-source data cannot enter the representation.

Concentration differs sharply between classes and matters for interpretation: PhiUSIIL legitimate averages 1.0207 URLs per U1, PhiUSIIL phishing 2.3199, the crawler-overlap pool 23.27 and the crawler-disjoint pool 20.85.

10. URL Representations

Code	Definition
R1	raw URL, unmodified
R2	path and query stripped
R3	<code>www</code> prefix plus path and query stripped
R4	fully normalised to <code>https://www.<registrable domain></code>

The chain is nested, so the reductions telescope exactly:

$$(R1 - R2) + (R2 - R3) + (R3 - R4) = R1 - R4$$

This identity is verified numerically inside every bootstrap replicate. The three components are referred to throughout as the **path/query**, **www prefix** and **residual host/subdomain** components of the observed reduction.

11. Experimental Design

The design is factorial over phishing source × benign population, evaluated at each of the four representations.

Cell	Composition	Designation
A	PhiUSIIL phishing + PhiUSIIL legitimate	primary
B	PhiUSIIL phishing + crawler-overlap benign	primary
B'	PhiUSIIL phishing + crawler-disjoint benign	sensitivity
C	PhishTank phishing + PhiUSIIL legitimate	primary
D	PhishTank phishing + crawler-overlap benign	primary
D'	PhishTank phishing + crawler-disjoint benign	sensitivity

Four contrasts are **primary** and were fixed before analysis:

1. **A vs C** — phishing source substituted, benign held at PhiUSIIL legitimate.
2. **B vs D** — phishing source substituted, benign held at crawler-overlap.
3. **The factorial interaction** (D–C) – (B–A).
4. **Platform-identity masking**.

Every other analysis in this report — B' and D', repeated-split stability, cross-source transfer, independent-benign transfer, U3, the temporal subset, and the real-world stress test — is secondary, sensitivity or exploratory, and is labelled as such where it appears. No family-wide significance is claimed beyond the four planned contrasts.

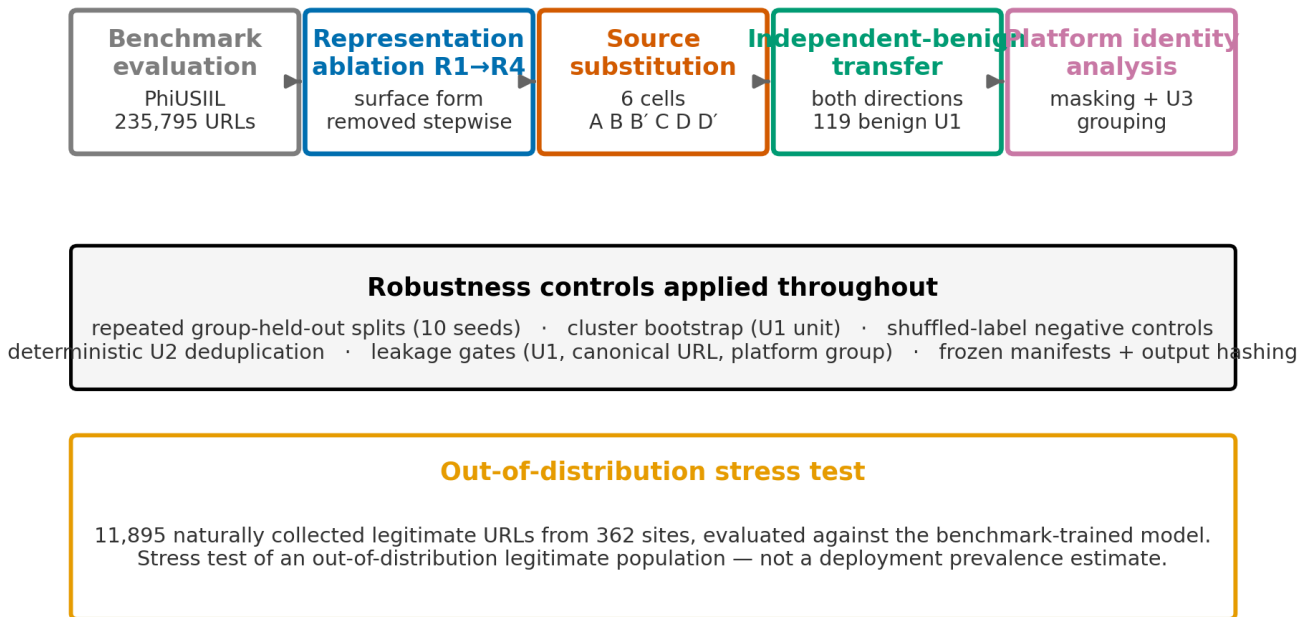


Figure 1. Study design of the evaluation. The benchmark is subjected in sequence to a nested representation ablation, source substitution across a six-cell factorial, independent-benign cross-source transfer, and platform-identity analysis, with robustness controls applied throughout and a separate out-of-distribution stress test. Schematic; no measured quantity is plotted.

12. Statistical Methodology

12.1 What receives statistical weight

Layer	Unit
Prediction	individual URL (one score per row)
Train/test grouping	U1 registrable domain
Bootstrap	U1 registrable domain
Metric weighting	per URL row — not domain-normalised

ROC-AUC is computed without sample weights, and PR-AUC at natural prevalence, accuracy, precision, TPR, FPR and TPR at fixed FPR are likewise unweighted at row level. The prevalence-reweighted PR-AUC applies a class-constant weight — each negative receives weight $n_{\text{pos}} / n_{\text{neg}}$ and each positive weight 1, giving an effective prevalence of 0.5. That reweighting rebalances the two **classes**; it does not equalise **domains**.

Consequently a domain contributes to a headline metric in proportion to how many URLs it carries, and that differs by more than twentyfold across benign populations: 1.02 URLs per U1 for PhiUSIIL legitimate against 23.27 for the crawler-overlap pool and 20.85 for the crawler-disjoint pool. In the crawler-benign cells a few hundred domains supply the entire benign class at roughly twenty URLs each. Cross-cell comparisons involving those cells are **URL-weighted comparisons and should be described as such**. No domain-level metric normalisation is applied anywhere in this study.

12.2 Uncertainty, reported at two levels and never pooled

Within-split. A cluster bootstrap over U1 units, 1,000 replicates, using multinomial cluster weights verified equivalent to index-based cluster resampling to $1e-12$. For representation analysis the same cluster draw is applied to R1–R4 simultaneously, so the components are **paired**. For cross-source comparisons independent draws are used where the cells do not share a test set.

Across-split. Ten repeated group-held-out splits with model refitting. This component is **not** captured by the bootstrap, and for normalised performance it is the larger of the two: cell C R4 has a bootstrap-implied standard deviation near 0.034 against an across-split standard deviation of 0.0607. Confidence intervals attached to single-split values therefore understate total uncertainty and must be read alongside §13.6.

Seeds vary the group-held-out partition only; model configuration, preprocessing, feature definitions and thresholds are fixed. The ten splits are draws from one finite dataset with overlapping training sets and are not independent replicates, so intervals computed across them describe split sensitivity rather than a sampling distribution for a new corpus.

12.3 Operating points

Fixed-FPR points use `searchsorted(fpr, target, 'right') - 1` with no interpolation, returning the largest attainable point whose achieved FPR is at or below nominal. Achieved FPR, threshold, negative count and the minimum resolvable FPR ($1 / n_{\text{neg}}$) are reported with every such value, and points resting on fewer than ten negatives are marked granularity-limited rather than presented as comparable.

13. Results

Each subsection follows the same order: question, setup, result, statistical uncertainty, interpretation, limitation.

13.1 Raw Benchmark Performance

Question. How high is measured performance on the benchmark under the conventional raw-URL representation?

Setup. XGBoost on 38 URL-string features; PhiUSIIL phishing against PhiUSIIL legitimate (cell A); single frozen domain-grouped split (`GroupShuffleSplit`, `test_size = 0.20`, `seed 42`, `groups = U1`).

Result. ROC-AUC 0.9988. Across all six cells raw-representation AUC lies between 0.9886 and 0.9998.

Uncertainty. Raw performance is exceptionally stable across repeated splits: the R1 standard deviation is 0.0002 in cell A and at most 0.0026 in any cell.

Interpretation. H1 is established. Near-ceiling raw performance is reproducible and is not an artifact of one split.

Limitation. A near-ceiling score constrains nothing about which properties of the URL produced it. That question is what §13.2 addresses.

13.2 Representation Sensitivity

Question. How much of the measured performance survives removal of URL surface form?

Setup. The nested chain $R1 \rightarrow R2 \rightarrow R3 \rightarrow R4$ applied identically to every cell, same model configuration and split throughout. The resulting ROC-AUC values are plotted in Figure 2 and tabulated in Table 2.

Result.

Cell	R1	R2	R3	R4	Designation
A	0.9988	0.9924	0.9640	0.7297	primary
B	0.9988	0.9873	0.9716	0.6794	primary
B'	0.9936	0.9328	0.9302	0.6381	sensitivity
C	0.9998	0.9897	0.9449	0.8633	primary
D	0.9928	0.9710	0.9229	0.8544	primary
D'	0.9886	0.9533	0.9479	0.6302	sensitivity

Full normalisation reduces AUC in every cell, to between 0.6302 and 0.8633.

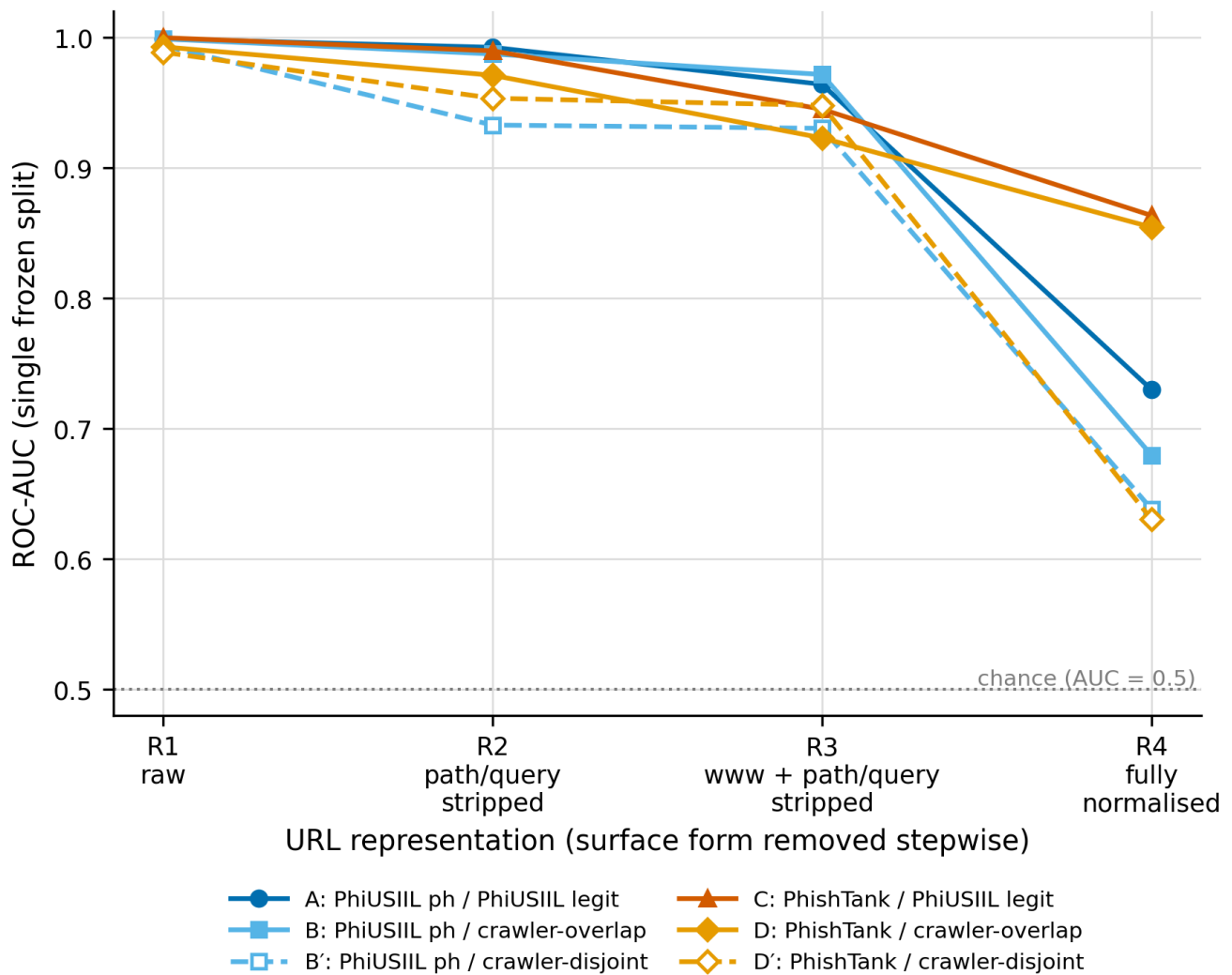


Figure 2. Representation sensitivity across the six P3 source-composition cells. ROC-AUC is shown for the nested URL representations R1 (raw), R2 (path/query stripped), R3 (www plus path/query stripped) and R4 (fully normalised). Values are the frozen primary split (GroupShuffleSplit, test_size = 0.20, seed 42, groups = U1), XGBoost, URL-weighted ROC-AUC. Cells A, B, C and D are primary; B' and D' are sensitivity cells, drawn with dashed lines and open markers. Single-split values; R4 is markedly split-sensitive (Figure 6).

Uncertainty. Cluster-bootstrap 95% intervals on R4 are given in Table 2. Across-split variability is substantially larger and is treated in §13.6.

Interpretation. H2 is strongly supported. Measured performance is representation-sensitive: the same data, the same model and the same split produce materially different scores depending only on how much URL surface form the representation retains.

Limitation. The single-split values are exact and reproducible but are draws from a distribution whose spread at R4 reaches 0.1164 (§13.6). They should not be quoted to more precision than that spread supports.

13.3 Representation Decomposition

Question. Which stage of the nested chain accounts for the observed reduction?

Setup. Paired cluster bootstrap, 1,000 replicates, same cluster draw across R1–R4, telescoping identity verified per replicate. The decomposition is plotted in Figure 3 and tabulated in Table 3.

Result. Bootstrap means of the components, and the residual host/subdomain share of the observed reduction with its interval:

Cell	path/query	www prefix	residual host	total	residual share	share 95% CI
A	0.0065	0.0285	0.2338	0.2688	0.870	[0.844, 0.897]
B	0.0116	0.0172	0.2815	0.3103	0.905	[0.834, 0.944]
B'	0.0568	0.0042	0.2865	0.3475	0.825	[0.771, 0.868]
C	0.0105	0.0465	0.0808	0.1378	0.563	[0.376, 0.774]
D	0.0227	0.0509	0.0678	0.1414	0.482	[0.343, 0.620]
D'	0.0393	0.0071	0.2944	0.3408	0.838	[0.489, 0.954]

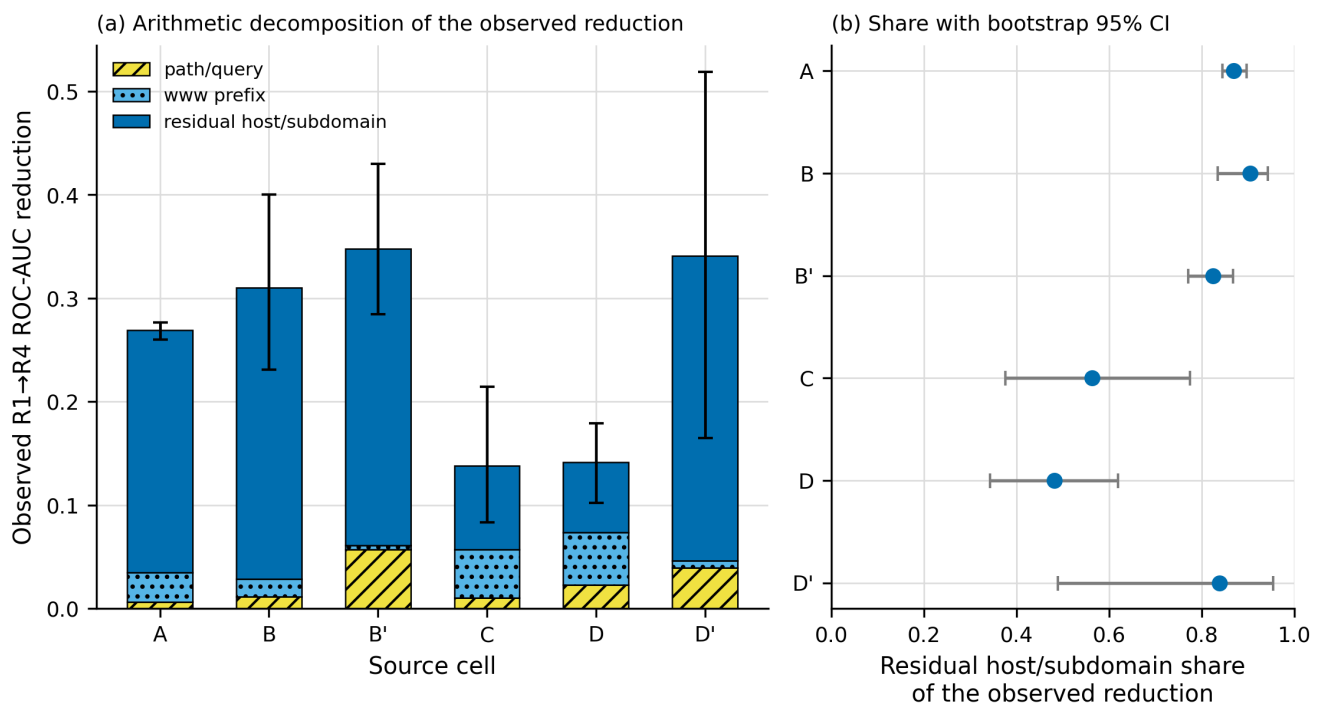


Figure 3. Decomposition of the observed R1-to-R4 ROC-AUC reduction into path/query, www-prefix and residual host/subdomain components, for all six cells. Panel (a) shows the arithmetic decomposition with a cluster-bootstrap 95% interval on the total; panel (b) shows the residual host/subdomain share with its bootstrap 95% interval. Values are paired U1-cluster bootstrap means over 1,000 replicates. Shares are shares of the observed raw-to-normalised AUC reduction; they are not percentages of model performance and not artifact proportions. The share intervals for cells C, D and D' are wide and the point share should not be read without them.

Uncertainty. The share is the bootstrap mean of per-replicate ratios, which is not the ratio of the component means. Its interval is narrow in the PhiUSIIL-phishing cells and wide in the PhishTank-phishing cells: cell C spans [0.376, 0.774] and cell D' spans [0.489, 0.954].

Interpretation. H3 is strongly supported in the PhiUSIIL-phishing cells, where residual host and subdomain structure accounts for the large majority of the observed reduction. In the PhishTank-phishing cells the share is lower and far less precisely estimated.

Terminology. These are **shares of the observed raw-to-normalised AUC reduction**. They are not shares of model performance, and they are not percentages of anything attributable to an artifact.

Limitation. The share is metric-dependent. At a fixed false-positive rate of 0.001 the residual host component ceases to dominate in the crawler-benign cells, where the path/query component becomes comparable or larger. No claim of universal residual-host dominance is made, and the point share should not be read without its interval.

13.4 Cross-Model Replication

Question. Is representation sensitivity a property of the data or of the learner?

Setup. Three model families on a common U1-deduplicated, U1-grouped split: XGBoost and logistic regression on the 38 URL-string features, and a character n-gram TF-IDF model with logistic regression that receives no engineered features at all. The three model families are compared in Figure 4.

Result.

Model	R1	R2	R3	R4	residual share
XGBoost	0.9985	0.9920	0.9553	0.7212	84.4%
Logistic regression	0.9982	0.9907	0.9513	0.6905	84.8%
Character n-gram TF-IDF	0.9990	0.9938	0.9734	0.8343	84.4%

Naming note. R3 here is `www_and_path_stripped`, matching the definition used throughout this report. The frozen file also contains a `www_stripped` field (www removed, path **retained**) whose value is 0.9969 for XGBoost; that field is a different representation and is **not** R3. It is not used anywhere in this report.

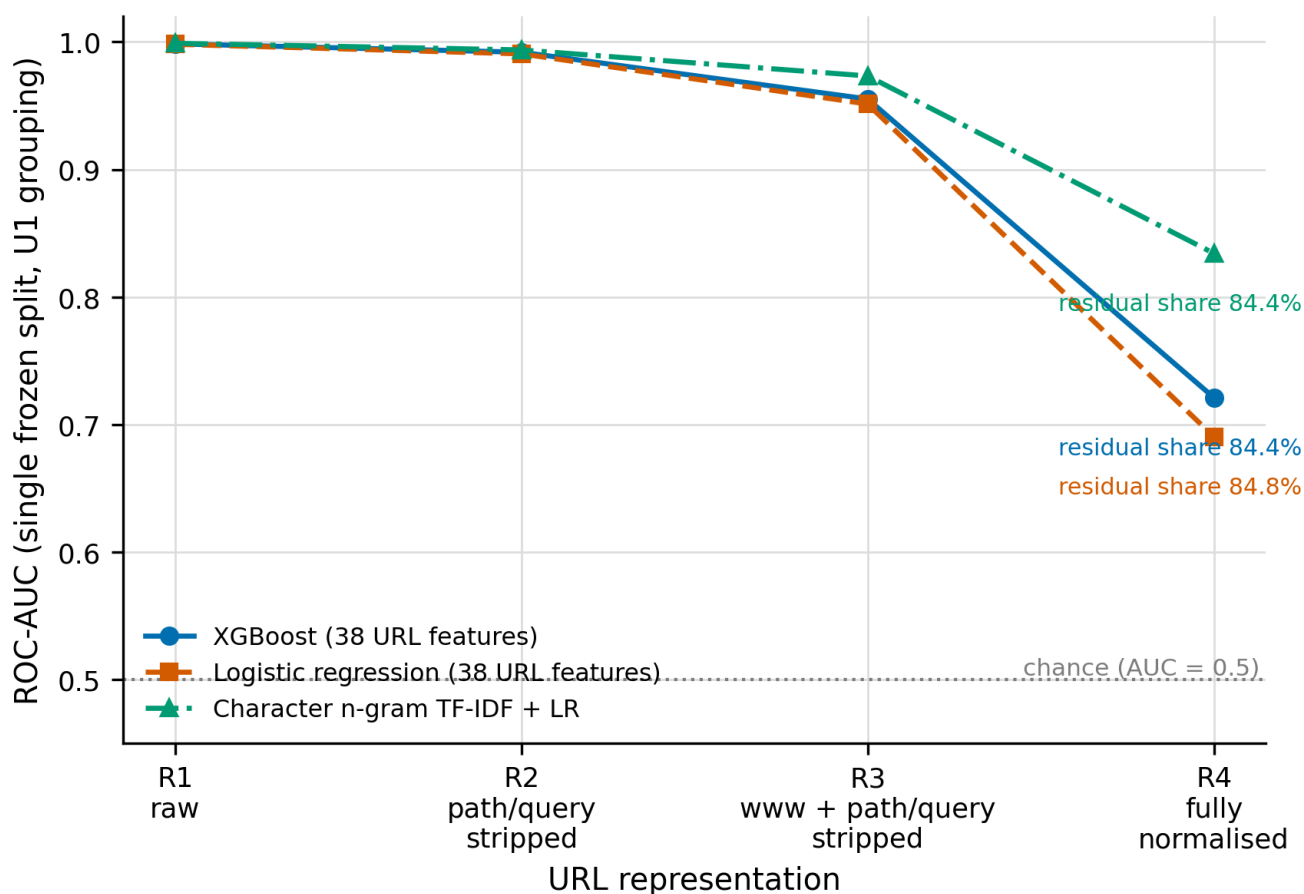


Figure 4. Representation sensitivity across three model families on a common U1-deduplicated, U1-grouped split: XGBoost and logistic regression on 38 URL-string features, and a character n-gram TF-IDF model with logistic regression that receives no engineered features. ROC-AUC against the nested representations R1-R4, where R3 is `www_and_path_stripped`. Annotated percentages are each model's residual host/subdomain share of the observed reduction. Single split per model; the purpose is replication of a pattern rather than precise estimation of any one value.

Uncertainty. Single split per model; the purpose is replication of a pattern, not precise estimation of any one value.

Interpretation. The effect is not specific to gradient boosting or to the engineered feature set. A model given only raw character n-grams shows the same qualitative collapse and a residual share within a percentage point of the other two.

Limitation. One dataset and one split. This establishes that the phenomenon is not learner-specific; it does not establish its magnitude in other corpora.

13.5 Cross-Source Evaluation

Question. Does normalised performance depend on which phishing corpus and which benign population constitute the evaluation set?

Setup. The six-cell factorial of §11, evaluated at R4. Contrasts are bootstrap means of the paired difference with 95% percentile intervals, from the frozen contrast table. Normalised performance by cell is plotted with its bootstrap intervals in Figure 5; the underlying values are in Table 2.

Result — primary contrasts.

Contrast	Estimate	95% CI	Interval excludes zero
A – C	-0.1333	[-0.1888, -0.0597]	yes
B – D	-0.1639	[-0.2599, -0.0734]	yes
B' – D' (<i>sensitivity</i>)	-0.0040	[-0.1982, +0.1906]	no

Result — factorial interaction. $(D - C) - (B - A) = +0.0413$, with a 95% interval including zero.

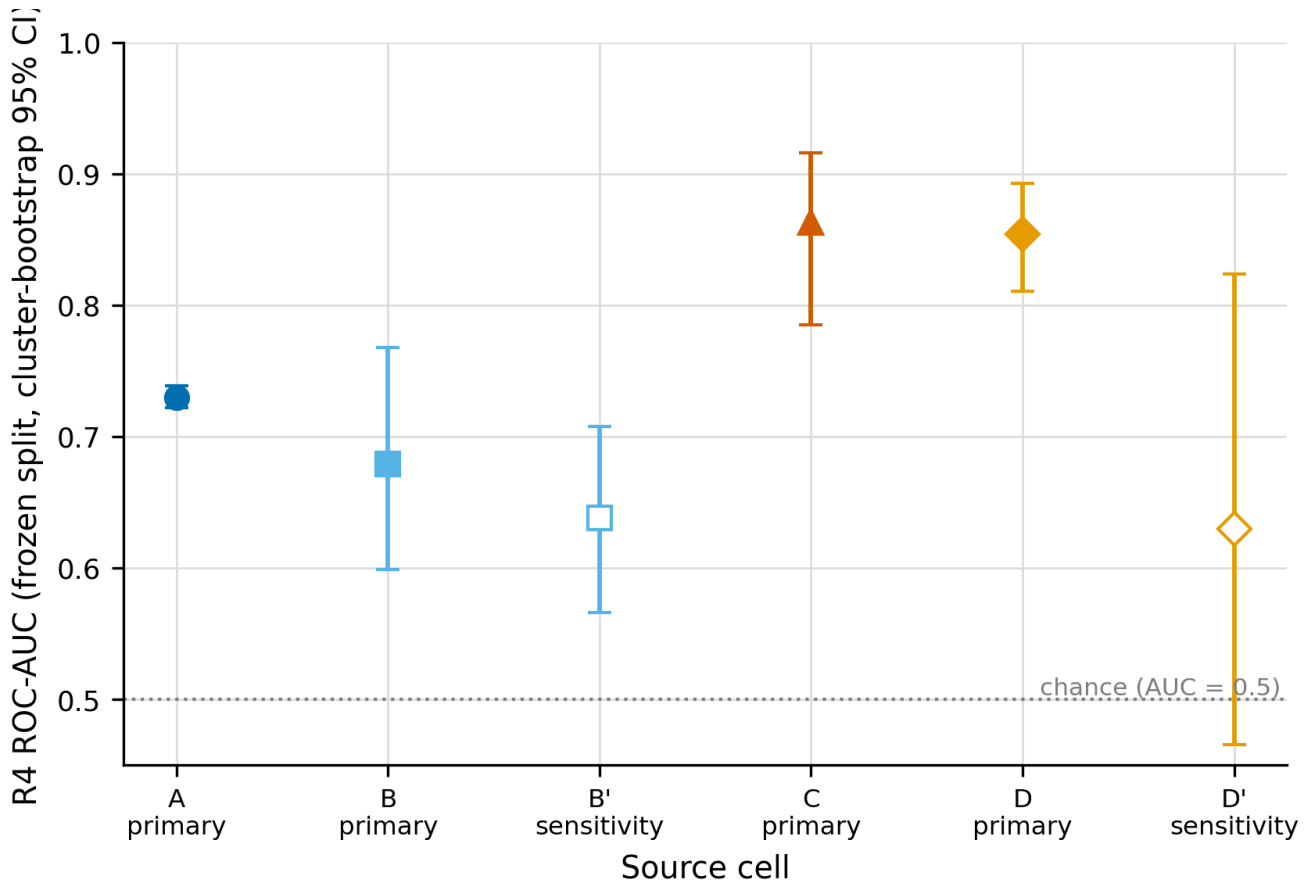


Figure 5. Normalised (R4) ROC-AUC by source composition for all six cells, with U1-cluster-bootstrap 95% intervals from the frozen primary split. Primary cells (A, B, C, D) and sensitivity cells (B', D') are labelled on the axis; no cell is omitted. Intervals capture test-set resampling only and exclude split-and-retrain variance. The two primary source contrasts have intervals excluding zero, while the four-cell interaction remains inconclusive.

Uncertainty. The two primary contrasts have intervals excluding zero. The sensitivity contrast B' – D' does not, so the effect is not uniform across benign populations.

Interpretation. H4 is strongly supported: substituting the phishing corpus while holding the benign population fixed moves normalised performance by roughly 0.13–0.16 AUC. H5 is **inconclusive**. The source-combination differences are strong, but the formal interaction does not establish differential dependence on the phishing-source versus benign-source factor. A null interaction is not evidence that no interaction exists; the design is not powered to support that claim.

Limitation. The cells share underlying data — cells A and C share their benign population, cells A and B share their phishing corpus — so contrasts across the factorial are not independent. Cell B versus B' additionally changes the domain population and the set of domains willing to be crawled at the same time, and is therefore not a pure population contrast.

13.6 Repeated-Split Stability

Question. How much of a single-split normalised value is attributable to the particular partition drawn?

Setup. Ten group-held-out splits per cell with model refitting, all four representations, model configuration fixed. The across-split distributions are plotted in Figure 6 and summarised in Table 4.

Result.

Cell	R4 median	R4 SD	R4 min	R4 max	R4 range	R1 SD	frozen single split
A	0.7484	0.0307	0.7175	0.8172	0.0996	0.0002	0.7297
B	0.7005	0.0619	0.5787	0.7508	0.1721	0.0012	0.6794
B'	0.6130	0.0419	0.5360	0.6871	0.1511	0.0021	0.6381
C	0.8730	0.0607	0.7289	0.9120	0.1831	0.0002	0.8633
D	0.8054	0.0645	0.6561	0.8717	0.2155	0.0013	0.8544
D'	0.7737	0.1164	0.5374	0.8709	0.3334	0.0026	0.6302

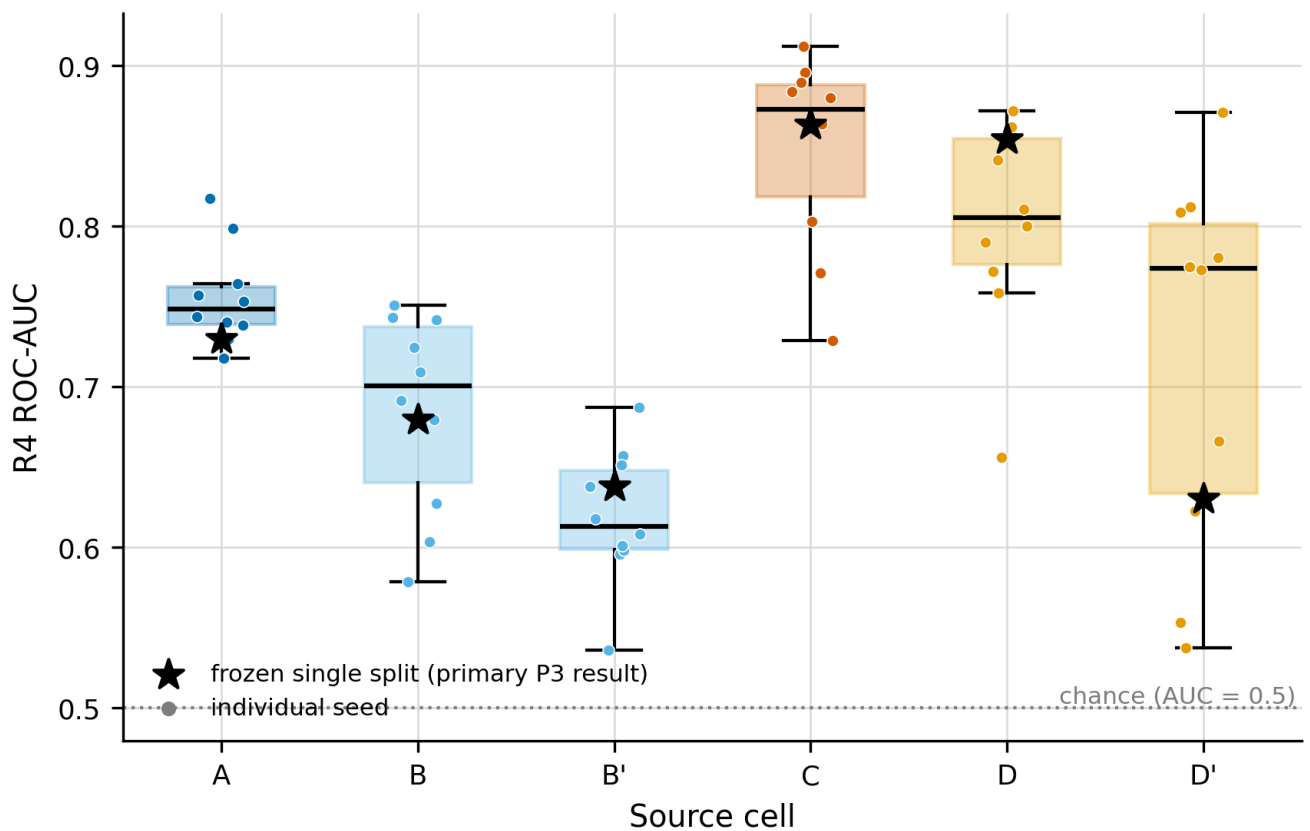


Figure 6. Stability of normalised (R4) ROC-AUC across ten repeated group-held-out splits per cell, with model refitting at each split. Boxes show the across-split distribution, points the individual seeds, and the star the frozen single split that constitutes the primary result. Seeds vary the partition only; model configuration, preprocessing and thresholds are fixed. The ten splits resample one finite dataset with overlapping training sets and are not independent replicates, so the distributions describe split sensitivity and are not population estimates.

Uncertainty. R4 standard deviation ranges from 0.0307 (cell A) to 0.1164 (cell D'). R1 standard deviation never exceeds 0.0026.

Interpretation. Normalised performance is markedly more split-sensitive than raw performance, by one to two orders of magnitude in standard deviation. The instability is a property of the normalised representation, not of the pipeline: under R4 every test domain is unseen and the model has only a registrable-domain string to work from.

Reporting rule. The frozen single-split values remain the primary results of the factorial. Seed medians characterise robustness and **do not replace them**; the two are not interchangeable and are never substituted for one another in this report.

Limitation. The ten splits are not independent replicates. They resample one finite dataset with heavily overlapping training sets.

13.7 Depth-stratified sensitivity analysis

Secondary / sensitivity analysis. Not a primary result.

Question. The source populations differ markedly in how deep their URLs reach into a site. Could the source-dependent variation reported in §13.5 be a depth effect rather than a source effect?

Setup. Depth is defined exactly as in the frozen analysis: `path_depth = count of non-empty '/'-separated path components from urlparse; query and fragment excluded`. Under that definition 100.0% of PhiUSIIL legitimate URLs sit at depth 0, against 72.8% of PhiUSIIL phishing, 38.8% of PhishTank, 0.0% of the crawler-overlap pool and 0.0% of the crawler-disjoint pool. The benign class of cells A and C is therefore entirely homepages, while the crawler-benign pools contain none, so depth is a plausible population-composition variable and is treated as one. Two complementary adjustments were applied: post-stratification reweighting within the depth strata a cell actually supports, using $w_i = q(d_i) / p_{\text{class}}(i)(d_i)$; q = pooled stratum share over overlapping strata; p_{class} = that class's within-support depth distribution, and an exact-matched variant that pairs phishing and benign URLs at identical depth. Weights use depth and class membership only and never use model predictions.

Result — depth-controlled R4. Values, deltas and cluster-bootstrap 95% intervals are read from the frozen depth-control outputs.

Cell	Original R4	Depth-controlled R4	Δ	95% CI	Exact-matched R4	Exact-matched n
A	0.7297	0.7378	+0.0081	[0.7300, 0.7461]	0.7404	16,568
B	0.6794	0.7395	+0.0601	[0.6953, 0.7801]	0.7426	4,250
B'	0.6381	0.6926	+0.0546	[0.6301, 0.7491]	0.6928	4,238
C	0.8633	0.8731	+0.0098	[0.7462, 0.9534]	0.8738	5,644
D	0.8544	0.7958	-0.0586	[0.7599, 0.8319]	0.8032	3,406
D'	0.6302	0.5163	-0.1139	[0.3714, 0.7689]	0.5692	3,190

Result — depth-controlled contrasts. Because the cells do not share a common depth support, contrasts are formed only within a regime both cells occupy:

Contrast	Regime	Estimate	95% CI	Interval excludes zero
A – C	depth=0 restricted	-0.1353	[-0.2156, -0.0064]	yes
B – D	depth>=1 reweighted	-0.0563	[-0.1138, -0.0046]	yes
B' – D'	depth>=1 reweighted	+0.1763	[-0.0983, +0.3320]	no

No four-cell interaction is formed. The frozen analysis records the reason: cells A and C are restricted to depth 0 because their benign source is entirely homepages, while B and D are restricted to depth ≥ 1 because the crawler-overlap pool contains none. No stratum is shared by all four cells, so no post-stratification weight built from the observed data can place them on a common depth support. Per-cell depth-controlled values are reported above so that the quantity can be formed by anyone who disagrees with that judgement.

Interpretation. Depth adjustment changes several normalised AUC estimates, but does not uniformly eliminate the observed source differences. The A–C comparison remains negative under homepage-only matching, while the B–D comparison remains negative when restricted to deeper URLs. These results weaken the interpretation that the source effects are explained solely by URL depth, but they do not fully deconfound site architecture, crawlability, or other correlated population differences.

Limitation. The adjustment is a reweighting of the observed data, not a randomised manipulation of depth. Retained fractions vary sharply between cells — the phishing class retains 77.4% of its rows in cell A but only 10.3% in cell B — so the depth-controlled cells are not the same populations as the unadjusted cells. Cell D' carries the widest interval ([0.3714, 0.7689]) and its point estimate should not be read on its own. This analysis is secondary and does not alter any primary contrast.

13.8 Independent-Benign Transfer

Question. Does normalised discrimination survive when **both** the phishing and the benign source are replaced?

Setup. A model is fitted on one phishing corpus plus a benign population and evaluated on the other phishing corpus plus a disjoint hold-out of that same independent benign population. The benign class is `crawler_disjoint`, which has zero U1 overlap with either phishing corpus. The 593 shared phishing domains are excluded from both corpora in both roles. Character n-gram TF-IDF with logistic regression; ten seeds; the external test population is fixed for a direction. External performance by representation and direction is plotted in Figure 7 and tabulated in Table 5.

Result. External normalised AUC, median over ten seeds:

Direction	R1	R2	R3	R4
PhiUSIIL → PhishTank	0.9543	0.8547	0.7896	0.6936
PhishTank → PhiUSIIL	0.9550	0.8044	0.8689	0.7061

Normalised discrimination remained above chance in **all 20 direction-seed combinations**, with an observed minimum of 0.6411.

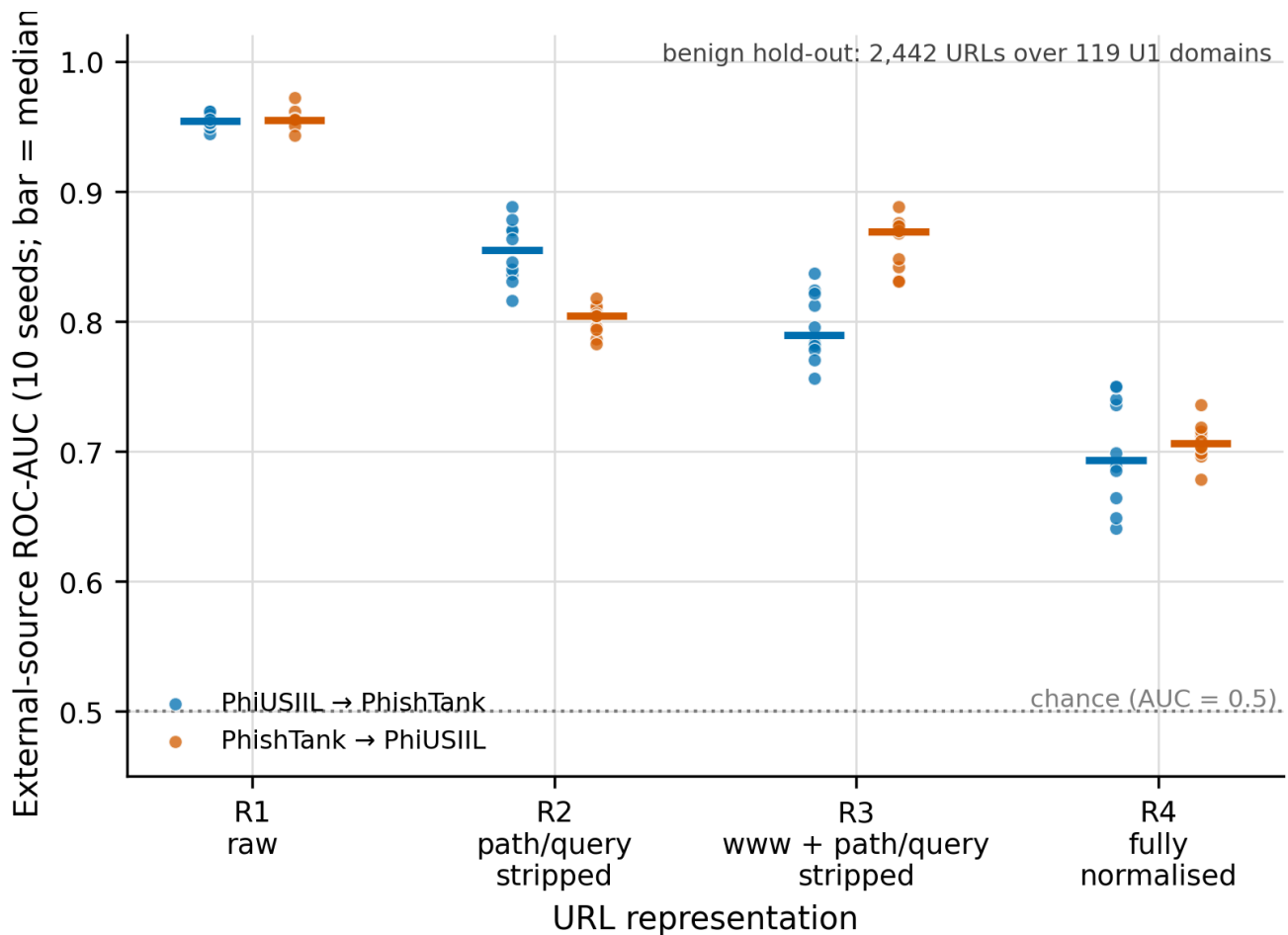


Figure 7. Cross-source transfer with a benign population independent of both phishing corpora. External-source ROC-AUC is shown by representation for both directions, ten seeds each, with the bar marking the median. The benign hold-out comprises 2,442 URLs over only 119 U1 domains, annotated on the panel. Domain-level disjointness is enforced between the training and external corpora. The result supports persistence of above-chance normalised discrimination outside the original benchmark; it does not establish that the surviving discrimination is phishing-specific information, and its magnitude is imprecise on this small domain base.

Uncertainty. External R4 standard deviation is 0.0416 and 0.0151 across the two directions.

Interpretation. H6 is supported with limitations. Above-chance normalised discrimination persists outside the original benchmark when both source populations change. This is consistent with something transferable surviving normalisation; it does **not** establish that the surviving discrimination is phishing-specific information.

Limitation — stated with the result, not deferred. The benign hold-out comprises 2,442 URLs over only **119 U1 domains**, drawn from a pool with a documented crawlability and popularity gradient. The minimum resolvable false-positive rate is $4.10e-04$, and at normalised representation the negatives collapse to roughly one distinct string per domain, so fixed-FPR operating points in this analysis are granularity-limited. The magnitude of the transfer estimate is correspondingly imprecise; the direction is the robust part.

Comparison to the shared-benign variant. When the benign class is instead held constant as PhiUSIIL legitimate, external R4 is approximately 0.803 and 0.806 — roughly 0.10 higher. In that variant raw-representation external AUC slightly *exceeded* its own same-source internal baseline, which is the signature of discrimination resting on the invariant side of the problem. The independent-benign values are the more conservative reading.

13.9 Platform Identity Analysis

Question. Does knowing *which* hosting platform a URL sits on help or hinder held-out normalised generalisation?

Setup. A frozen platform list (18 services, 24 domains, externally sourced and frozen before any prevalence figure was computed) defines recognised platforms. Masking replaces a recognised platform's registrable domain inside the authority component with a neutral token, leaving the host present: all platforms become mutually indistinguishable while remaining distinguishable from non-platform hosts. Ten seeds per cell; the comparison is paired within each seed. Effect is defined as **AUC(unmasked) – AUC(masked)**, so a **negative** value means masking increased AUC. The paired comparison and the resulting effect are plotted in Figure 8; per-cell statistics are in Table 6.

Result.

Cell	mean effect	median	SD	min	max	negative / positive
A	-0.0546	-0.0447	0.0376	-0.1477	-0.0156	10 / 0
B	-0.1444	-0.1337	0.0989	-0.2757	-0.0367	10 / 0
B'	-0.1531	-0.1497	0.0772	-0.3134	-0.0536	10 / 0
C	-0.0656	-0.0436	0.0577	-0.1833	-0.0172	10 / 0
D	-0.1060	-0.0816	0.0721	-0.2527	-0.0182	10 / 0
D'	-0.1655	-0.0900	0.1321	-0.3929	-0.0084	10 / 0

All **60 of 60** seed-cell comparisons were negative. The independent within-split cluster bootstrap excluded zero in 53 of those 60 splits.

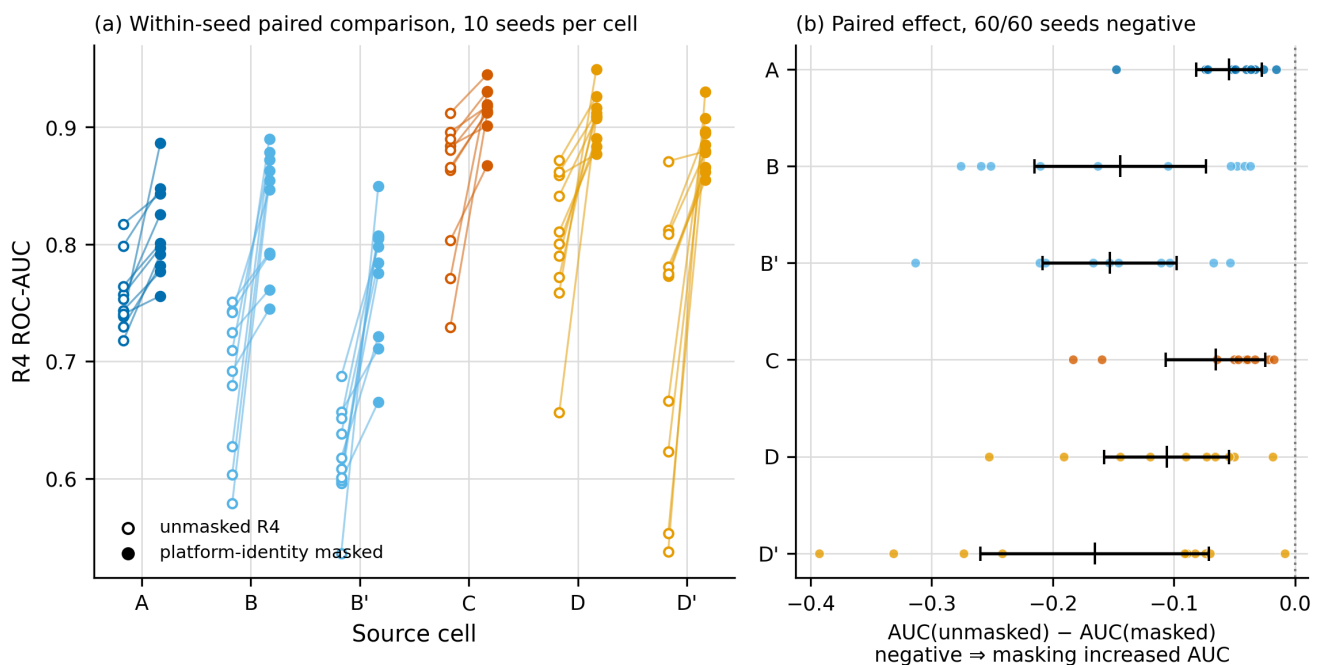


Figure 8. Effect of masking specific hosting-platform identity on normalised (R4) ROC-AUC. Panel (a) shows the within-seed paired comparison of unmasked against masked for ten seeds in each of the six cells; panel (b) shows the paired effect, defined as AUC(unmasked) minus AUC(masked), so a negative value means masking increased AUC. All 60 of 60 seed-cell comparisons are negative. The manipulation neutralises which platform a host belongs to while leaving the host present: it tests platform identity, not platform membership, and does not imply that platform membership is uninformative.

Uncertainty. Two uncertainty sources agree rather than conflict: the across-seed sign is unanimous, and the within-split paired bootstrap independently excludes zero in the large majority of splits.

Interpretation. Masking specific platform identity consistently increased held-out normalised AUC, which is causal with respect to the masking manipulation. This is consistent with the model using specific platform identity in a way that does not transfer to unseen registrable domains. Note the direction: it is opposite to the pre-registered expectation.

Boundary — identity is not membership. The manipulation neutralises *which* platform a host belongs to; it does not remove the fact that the host is a platform. Nothing here implies that platform membership is uninformative, and no such claim is made.

Limitation. The frozen platform list carries a disclosed circularity risk: it was written after the project had observed PhiUSIIL's most frequent phishing platforms. The mitigation is that every entry is justified by an external published source predating the project, and entries coinciding with observed frequencies are flagged for transparency.

13.10 U3 Platform-Grouped Evaluation

Question. Does normalised performance degrade when **entire platform groups** are held out of training?

Setup. U3 = U2 deduplication plus platform-grouped splitting, where recognised platform domains are grouped by service identity and unrecognised domains are grouped individually. Ten matched seeds against the U2 baseline, cells A, C and D.

Result. U3 normalised medians: A 0.7486, C 0.8425, D 0.8100. Across ten matched seeds, all twelve cell × representation mean differences had confidence intervals including zero. No systematic U2 → U3 degradation was detected.

Uncertainty. The delta distributions have standard deviations comparable to the already-measured split-draw noise floor, so the comparison has limited power to detect a small systematic effect.

Interpretation — INCONCLUSIVE. Only **15 recognised platform groups** appear in these cells, and in some seeds the test fold contains none of them at all. U1 grouping already prevents subdomain-level platform sharing, so U3's incremental reach was three to four merged groups out of roughly 150,000.

This result does not establish the absence of platform leakage, the absence of platform effects, or equivalence between U2 and U3. It establishes that with this many platform groups, no systematic difference was detectable.

13.11 Real-World Stress Test

Question. How does a benchmark-trained model behave on a legitimate population collected by a different process?

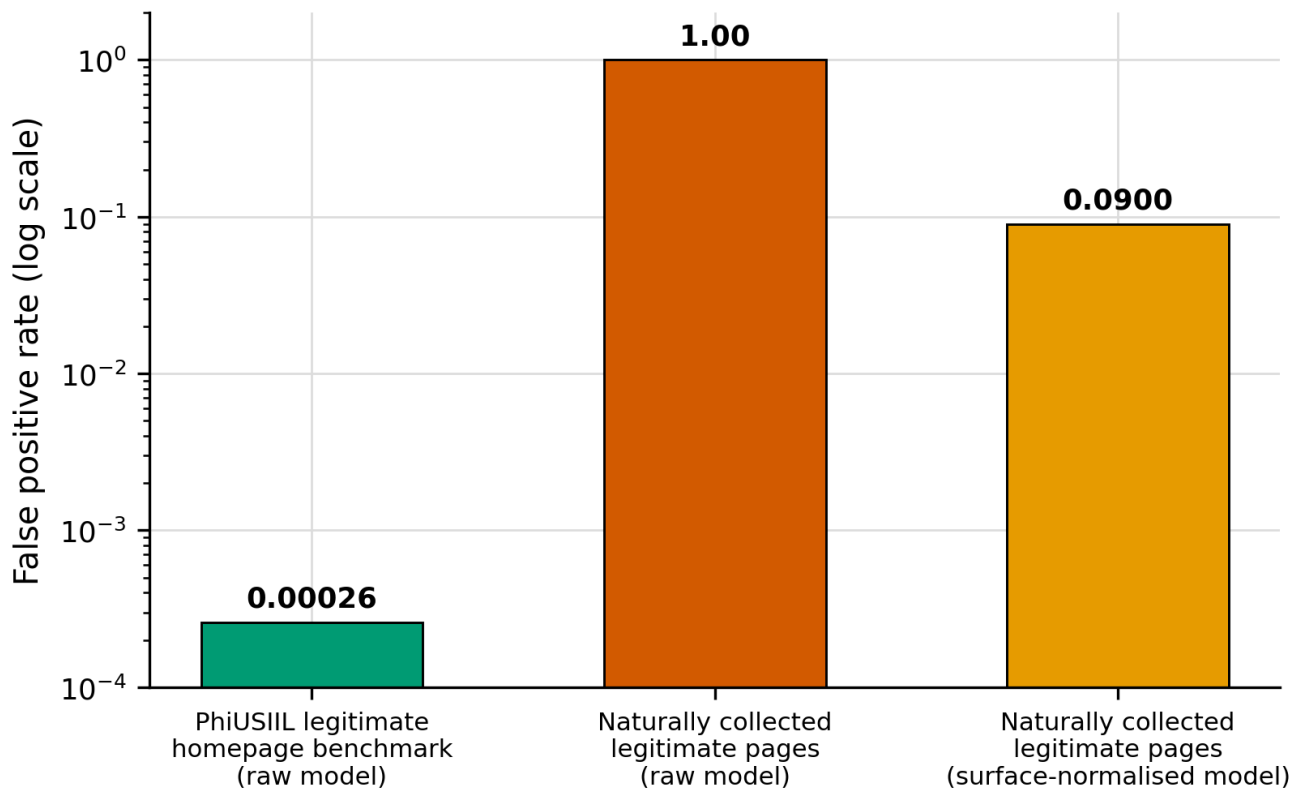
Setup. 11,895 naturally collected legitimate URLs from 362 sites, evaluated against a model trained on the benchmark. Benchmark reference is the **held-out test split (raw model, $\tau = 0.5$)**. The comparison is plotted in Figure 9.

Result.

Population	Model	FPR
PhiUSIIL legitimate, held-out test split	raw	0.00026 (7 / 26,946)
Naturally collected legitimate pages	raw	1.0000

Population	Model	FPR
Naturally collected legitimate pages	surface-normalised	0.08995

Median per-site flag rate under the raw model is 1.000.



11,895 URLs from 362 sites. Stress test of an out-of-distribution population; not a deployment prevalence estimate and not a representative real-world false-positive rate.

Figure 9. False-positive rate of a benchmark-trained model on two legitimate populations, log scale. The benchmark reference is the held-out test split (raw model, $\tau = 0.5$), $7 / 26,946 = 0.00026$. The naturally collected population comprises 11,895 URLs from 362 sites, evaluated under the raw model and under the surface-normalised model. This is an out-of-distribution stress-test observation: it is not a deployment prevalence estimate and the value of 1.0 must not be read as a representative real-world false-positive rate.

Uncertainty. Single evaluation on a fixed harvested population; no repeated sampling.

Interpretation. This is an **out-of-distribution stress-test observation**. A model that is near-perfect on the benchmark's legitimate class flags essentially the entire naturally collected legitimate population, and the surface-normalised model does not. The benchmark's legitimate class is bare homepages; the harvested population is not.

Boundary. The value 1.0 is **not** a deployment false-positive rate and must not be read as one. It is the behaviour of a benchmark-trained model on an out-of-distribution population, and the harvested population is neither a random sample of the web nor a prevalence-representative deployment stream.

13.12 Negative Controls

Question. Can the evaluation pipeline manufacture discrimination from randomised labels?

Setup. Labels shuffled at URL level and at cluster level, normalised representation, multiple seeds, otherwise identical pipeline. The control distributions are plotted in Figure 10 and tabulated in Table 7.

Result. URL-level shuffled-label R4 AUC:

Cell	mean	SD	min	max
A	0.5012	0.0034	0.4971	0.5063
B	0.5000	0.0056	0.4939	0.5061
B'	0.4956	0.0031	0.4925	0.5007
C	0.4999	0.0037	0.4940	0.5034
D	0.4937	0.0047	0.4886	0.4990

Cluster-level shuffling is near chance on average. No condition was flagged under the pre-specified rule $| \text{mean AUC} - 0.5 | > 0.05$ (0 flagged).

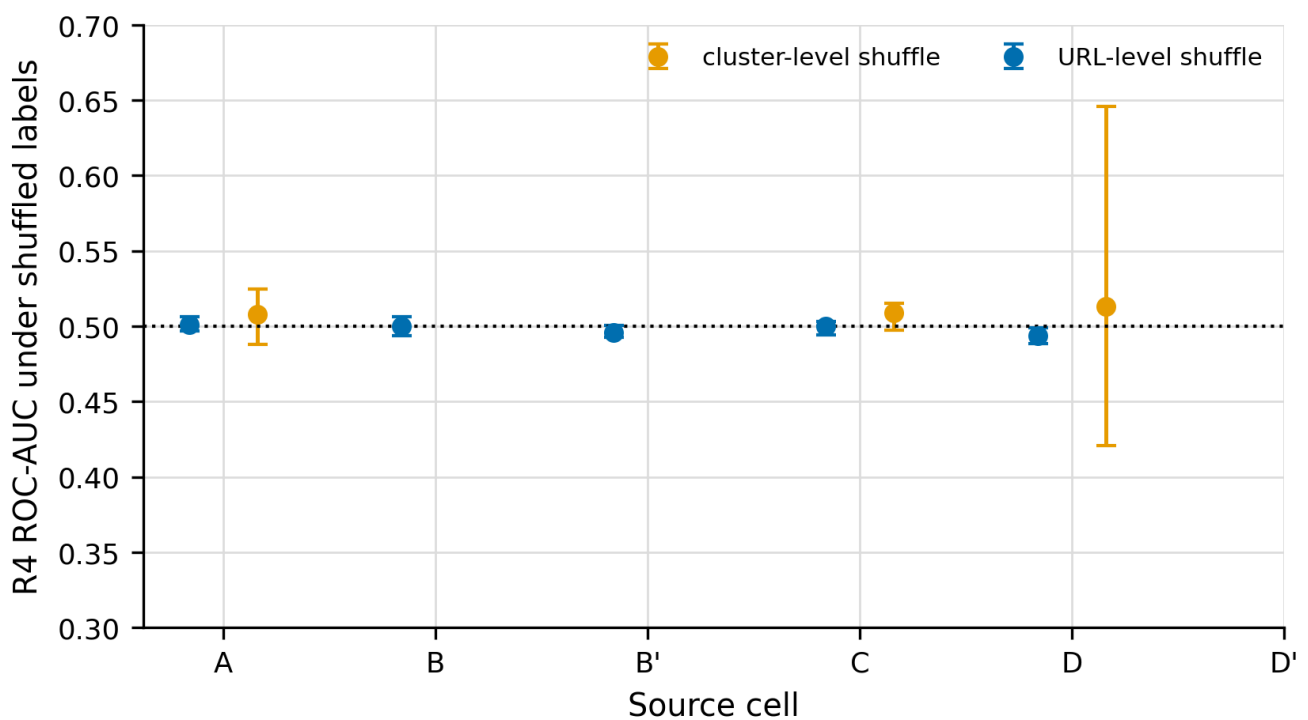


Figure 10. Shuffled-label negative controls at normalised representation. Mean ROC-AUC with the observed across-seed range is shown for URL-level and cluster-level label shuffling in each cell. Randomised labels produce approximately chance discrimination. This is pipeline-validity evidence only: it establishes that the evaluation pipeline does not manufacture discrimination from random labels, and carries no implication for the causal origin of the residual normalised signal.

Interpretation. The pipeline does not manufacture strong discrimination from randomised labels. This is **pipeline-validity evidence only**. It carries no implication for any scientific hypothesis about the origin of the residual normalised signal.

13.13 Temporal Sensitivity

Question. Do the cross-source findings depend on PhishTank's recency profile?

Setup. Secondary cells restricted to PhishTank entries submitted in 2023 or earlier. The feed is online-valid — currently verified live, not recently submitted — with median submission 2026-01 and 11.37% at or before 2023.

Result. Normalised AUC: C_le2023 0.6718, D_le2023 0.6945, D'_le2023 0.6291.

Provenance note — stated explicitly. These cells come from the same frozen factorial execution as the primary cells and therefore predate the case-normalisation correction to the U1 grouping key. A post-hoc audit established that all three temporal cells contain **zero case-colliding U1 values** (C_le2023 132,934 groups pre- and post-fix; D_le2023 1,802; D'_le2023 1,453; merged groups 0, rows affected 0). Because the group set is identical under both conventions, the partition, the fitted model and the resulting metrics are identical. These values are therefore **final-pipeline-equivalent**, and are reported as **secondary**.

Interpretation. Restricting PhishTank to older entries lowers normalised performance in the PhiUSIIL-legitimate and crawler-overlap cells relative to the full feed, consistent with a temporal component to the cross-source difference.

Limitation. The subset is small and the comparison is not a controlled temporal manipulation: it changes the phishing population's age and its size simultaneously.

14. Integrated Findings

Read together, the experiments support four statements and leave one central question open.

F1 — Measured performance is representation-sensitive. Raw-URL AUC of 0.9988 falls to 0.7297 under full normalisation in the benchmark cell, and the pattern replicates across three model families including one with no engineered features. Direct experimental finding.

F2 — Most of the observed reduction is associated with residual host and subdomain structure in the PhiUSIIL-phishing cells, and substantially less in the PhishTank-phishing cells, where the interval is wide. Associational, and metric-dependent.

F3 — Normalised performance is source-dependent. Substituting the phishing corpus moves R4 by roughly 0.13–0.16 AUC in the two primary contrasts, with intervals excluding zero. The formal factorial interaction is inconclusive, so the study does not establish which factor dominates. Direct experimental finding for the contrasts; inconclusive for the interaction.

F4 — Normalised performance is split-sensitive, with across-split standard deviation up to 0.1164 against at most 0.0026 for raw. Direct experimental finding.

Open question. Above-chance normalised discrimination persists when both source populations are replaced, and masking specific platform identity improves rather than degrades held-out performance. Together these constrain but do not identify the origin of the residual normalised signal.

15. Discussion

The findings are best read as a statement about measurement rather than about phishing. Three evaluation choices that are usually left implicit — the URL representation, the source composition of both classes, and the particular domain-grouped split — each moved the headline number in this study by more than the margins typically reported between competing detection methods. A benchmark AUC that does not state all three is therefore under-specified.

One obvious candidate explanation for the source differences — that they merely reflect differing URL path depth between the populations — was tested directly and does not account for them: the principal contrasts persist within the directly comparable depth strata (§13.7), though depth remains correlated with site architecture and collection mechanism, so that analysis is a sensitivity check rather than a deconfounding.

The residual normalised signal is the interesting open case. Two observations pull in different directions. Above-chance discrimination survives replacement of both the phishing and the benign source, which is what one would expect if some transferable property of phishing URLs were being captured. But masking specific hosting-platform identity *improves* held-out performance, which is what one would expect if the model were partly keying on identity information that does not generalise across domains. Both can be true simultaneously, and the design cannot separate them.

The results are consistent with shortcut-like source-dependent structure, but the study does not identify a specific causal shortcut mechanism. That qualification is not rhetorical hedging: no experiment here manipulates a candidate mechanism and observes the predicted change. Identifying one would require evaluation data with independently controlled collection processes, which this study does not have.

It is worth being explicit about what is *not* implied. Nothing here shows that PhiUSIIL is unsuitable for its intended purpose, that its construction is defective, or that published results using it are wrong. The benchmark behaves exactly as a benchmark built by combining two separately collected populations would be expected to behave. The methodological point is about how such scores should be reported and interpreted, not about the datasets themselves.

16. Threats to Validity

Threats are grouped by what they endanger. This distinction matters: several threats affect the **magnitude** of reported effects without endangering the **direction** of the headline finding.

16.1 Threats to the magnitude of reported effects

URL-versus-domain weighting. Headline metrics weight URLs equally while domains contribute unequal numbers of URLs — approximately 1.02 URLs per U1 for PhiUSIIL legitimate against 23.27 and 20.85 for the crawler pools. In the crawler-benign cells a few hundred domains supply the entire benign class. Grouping and bootstrap units are domain-level, so no domain crosses a split and clustering is respected in the uncertainty estimate, but cross-cell comparisons involving those cells are URL-weighted.

Crawlability and popularity bias. The disjoint pool succeeded on 605 of 1,200 queried domains against 946 of 1,200 for the overlap pool, and is lower-ranked by popularity. The B versus B' contrast therefore changes the domain population and the set of crawlable domains simultaneously and is not a pure population contrast.

Small independent-benign hold-out. 119 U1 domains in the fixed hold-out, 596 in the pool. Repeated splits resample a thin base and the transfer magnitude is correspondingly imprecise.

Split sensitivity. Normalised performance moves materially with the partition; single-split values are exact but are draws from a distribution with standard deviation up to 0.1164.

Finite FPR resolution. Where the negative pool is small, a nominal FPR of 0.001 corresponds to a handful of negatives, and at normalised representation the negatives collapse to roughly one distinct string per domain. Those operating points are reported with achieved FPR and negative counts and marked granularity-limited.

Cross-cell covariance. The six cells share underlying data — A and C share a benign population, A and B share a phishing corpus — so contrasts are not mutually independent and should not be treated as a family of independent tests.

Unit-definition dependence. Results depend on the registrable-unit definition used for grouping. U1, U2 and U3 give different group counts and therefore different partitions; the study reports the definition used for every number rather than assuming one canonical choice.

URL-depth composition. The source populations differ substantially in URL path depth: PhiUSIIL legitimate URLs are concentrated at the homepage level, whereas PhishTank and the crawler-benign populations contain substantially deeper URLs. A depth-stratified sensitivity analysis was therefore performed (§13.7). The principal source differences persisted in the directly comparable depth strata, although several normalised AUC estimates changed materially. Because depth is correlated with site architecture, crawlability, and collection mechanism, this analysis does not constitute full causal deconfounding.

16.2 Threats to the direction of the headline finding

Unknown PhiUSIIL provenance. The collection mechanism of the legitimate class is not documented at a verifiable level. This limits mechanistic explanation but does not affect the measured representation and source sensitivity, which are properties of the data as distributed.

PhishTank temporal mismatch and survivorship. The online-valid feed is biased toward currently live phishing, with median submission 2026-01. Long-lived phishing is over-represented relative to a random sample of reported phishing. The temporal sensitivity analysis (§13.13) probes this; it does not remove it.

Possible residual contamination in the crawled benign pools. A structural audit reviewed 200 URLs and found 0 visibly suspicious, Wilson 95% upper bound 0.019. The audit inspected URL strings only; no URL was visited. **It does not establish benign purity** and by construction understates true contamination.

Limited U3 platform-group power. 15 recognised groups, with some test folds containing none. The U3 null is uninformative about platform-group generalisation rather than evidence against it.

Null factorial interaction. The interaction interval includes zero. This constrains claims about which source factor dominates. It does not undermine the two primary contrasts, whose intervals exclude zero.

None of the threats in §16.2 plausibly reverses the direction of F1, F3 or F4, each of which is a within-study comparison under a fixed protocol. They bound how far those findings generalise beyond the corpora studied.

17. Interpretation Boundaries

The following are **not** established by this study and are not claimed:

- that the residual normalised signal reflects phishing-specific information;
- that the residual normalised signal is entirely attributable to source-dependent structure;
- that any specific percentage of model performance is attributable to an artifact — decomposition percentages refer only to the share of the *observed raw-to-normalised AUC reduction*;
- that a specific construction mechanism in PhiUSIIL produces the observed effects;
- that PhiUSIIL is unsuitable, defective, or invalid;
- that platform membership is uninformative — the masking manipulation tests platform *identity* only;
- that U3 demonstrates the absence of platform leakage or equivalence between grouping schemes;
- that R4 represents an artifact-free performance ceiling;
- that the null factorial interaction demonstrates the absence of an interaction;
- that the false-positive rate observed on the naturally collected legitimate population is a representative deployment rate.

18. Contributions

The contribution is a rigorous empirical evaluation framework showing that phishing-URL benchmark performance is substantially sensitive to URL representation and source composition. It is not a new classifier, a claim to have discovered universal phishing features, an identification of an artifact mechanism, or a separation of signal from artifact.

The methodological contribution is the **combination** of the following, applied to one corpus family under a single frozen protocol:

1. A nested representation ablation with an exact telescoping decomposition of the observed AUC reduction and paired cluster-bootstrap intervals.
2. A factorial source-substitution design that separates the phishing-source factor from the benign-population factor, with pre-specified primary contrasts.
3. Independent-benign cross-source transfer in both directions, with enforced domain-level disjointness and a measurement of the inflation produced by omitting it.
4. Repeated domain-grouped splits, quantifying a variance component that the conventional cluster bootstrap omits and which dominates it for normalised performance.
5. A platform-identity masking manipulation that separates platform identity from platform membership, with a frozen externally-sourced platform list.
6. Shuffled-label negative controls at URL and cluster level as pipeline validation.
7. An explicit audit and provenance chain in which every published value traces to a hashed frozen result file.

19. Implications for Benchmark Design

These are methodological recommendations grounded in the evidence above. They are not claims that existing benchmarks are invalid.

1. **Report multiple URL representations**, not raw URLs alone. A single representation does not characterise a detector's dependence on surface form.
2. **Report repeated domain-grouped splits**, not one split. Normalised performance in this study varied across splits by more than the differences usually claimed between methods.
3. **Include cross-source validation**. Train on one corpus and evaluate on an independently collected one, with domain-level disjointness enforced and verified.
4. **Include an independently collected benign population**. Holding the benign class fixed permits benign-side structure to carry discrimination.
5. **Report URL count and independent-domain count separately**, and state which unit carries statistical weight in each metric.
6. **Consider platform or service grouping** where hosting platforms are a material fraction of either class, and report how many such groups exist.
7. **Publish exact preprocessing, canonicalisation and grouping definitions**. Group-set changes as small as three merged groups re-randomise an entire partition under seeded group splitting.
8. **Use negative controls** for any evaluation pipeline complex enough that a leakage path is not obvious by inspection.
9. **Distinguish in-distribution benchmark performance from out-of-distribution stress tests**, and label the latter as such.

10. Avoid treating one benchmark AUC as a universal deployment estimate.

20. Reproducibility and Audit Trail

Every figure and table referenced here is generated by a builder script that reads frozen result files at run time; no published value is hand-transcribed. Frozen experimental outputs were hashed before and after every publication build and verified unchanged. A machine-readable artifact manifest records, per artifact, its source files, source metric, primary/exploratory designation and interpretive limitations. See §26 for the provenance summary and §25 for the frozen file references.

21. Conclusion

Measured phishing-URL detection performance is substantially representation- and source-dependent. Near-ceiling raw benchmark performance (0.9988 in the benchmark cell) degrades under progressive normalisation (0.7297); the effect replicates across three model families; normalised discrimination varies materially across source combinations; normalised performance is markedly more split-sensitive than raw performance; and a benchmark-trained raw model flags essentially the entire naturally collected legitimate population used as an out-of-distribution stress test.

Above-chance normalised discrimination persists when both the phishing and the benign source are replaced, on a small independent benign base, while masking specific hosting-platform identity improves rather than degrades held-out performance. These constrain the origin of the residual normalised signal without identifying it.

The study establishes benchmark sensitivity, not the unique causal identity of the residual signal.

22. Figure Inventory

Figures are numbered by first appearance in the body. The artifact ID is retained for traceability to `results/publication/figures/`.

Figure	Artifact ID	Section	Content
Figure 1	F01	§11	Evaluation-methodology study design
Figure 2	F02	§13.2	Representation sensitivity across the six source cells
Figure 3	F03	§13.3	Decomposition of the observed raw-to-normalised AUC reduction
Figure 4	F10	§13.4	Representation sensitivity across three model families
Figure 5	F05	§13.5	Normalised performance by source composition, with bootstrap CIs
Figure 6	F04	§13.6	Stability of normalised performance across ten repeated splits
Figure 7	F06	§13.8	Cross-source transfer with an independent benign population
Figure 8	F07	§13.9	Masking specific hosting-platform identity
Figure 9	F09	§13.11	Out-of-distribution stress test on naturally collected pages
Figure 10	F08	§13.12	Shuffled-label negative controls

Figures are embedded from title-free variants in `results/publication/figures/embedded/`, generated by re-executing the identical plotting code over the identical frozen inputs with the in-image title suppressed so that the caption carries it. The originals in `results/publication/figures/` are unmodified and remain available as vector PDF plus 300 DPI PNG.

23. Table Inventory

ID	Artifact	Content
T01	<code>T01_dataset_inventory</code>	Source inventory; URL and U1/U2 domain counts reported separately
T02	<code>T02_primary_p3_auc</code>	Primary P3 AUC by cell and representation, with R4 bootstrap CI
T03	<code>T03_representation_decomposition</code>	Decomposition components and shares with bootstrap CIs
T04	<code>T04_r4_split_stability</code>	R4 median, SD, min, max, range across 10 splits
T05	<code>T05_cross_source_transfer</code>	Cross-source transfer, both directions, both benign variants
T06	<code>T06_platform_masking</code>	Platform-identity masking effects with sign consistency
T07	<code>T07_negative_controls</code>	Shuffled-label negative controls
T08	<code>T08_claims_evidence_matrix</code>	Claims, evidence, status, causal level, limitations
Table 9	<code>T_depth_control</code>	Depth-controlled R4 by cell, with deltas, bootstrap CIs and exact-matched values (secondary)

Tables 1-8 are in `results/publication/tables/` as CSV, Markdown and LaTeX. Table 9 is reproduced in §13.7 from the frozen secondary output `results/P3_depth_control/T_depth_control.csv`.

24. Final Claims Matrix

#	Claim	Evidence	Status	Causal level	Main limitation
1	Raw benchmark performance is extremely high	Cell A R1 = 0.9988; all cells 0.9886–0.9998 (T02, F02)	Established	Descriptive	Says nothing about which URL properties produced it
2	Performance is representation-sensitive	R1→R4 reduction in all six cells; replicated in three model families (T02, T03, F02, F10)	Strongly supported	Causal w.r.t. the representation manipulation	Single-split primary values; R4 is split-sensitive
3	Most of the observed AUC reduction is associated with residual host/subdomain structure	Residual share 0.870 (A), 0.905 (B); lower and wide in C/D (T03, F03)	Strongly supported in PhiUSIIL-phishing cells	Associational	Share of the <i>observed reduction</i> only; metric-dependent; wide CIs in C, D, D'
4	Representation sensitivity is not XGBoost-specific	XGBoost, logistic regression and character n-gram all show it; residual shares ≈84% (F10)	Supported	Descriptive, replicated across learners	One dataset, one split
5	Normalised performance varies across source combinations	A–C -0.1333 [-0.1888, -0.0597]; B–D -0.1639 [-0.2599, -0.0734]	Strongly supported	Association under source substitution	B'–D' includes zero; factorial interaction inconclusive

#	Claim	Evidence	Status	Causal level	Main limitation
6	Independent-benign transfer remains above chance	R4 0.6936 and 0.7061; above chance in 20/20 direction-seed combinations, min 0.6411 (T05, F06)	Supported with limitations	Descriptive generalisation	119 benign hold-out U1 domains; crawlability bias; magnitude imprecise
7	Specific platform identity is not required for the observed normalised discrimination	Masking identity increased AUC in 60/60 seed-cell comparisons (T06, F07)	Strongly supported	Causal w.r.t. the masking manipulation	Tests identity, not membership; frozen list has disclosed circularity risk
8	Platform-grouped (U3) evaluation shows degradation	All 12 cell × representation mean deltas have CIs including zero	Inconclusive	Null / underpowered	15 platform groups; some test folds contain none; does not establish absence of leakage
9	Raw benchmark performance fails on naturally collected legitimate pages	FPR 0.00026 held-out benchmark vs 1.0000 on 11,895 URLs from 362 sites (F09)	Supported	Out-of-distribution stress test	Not a deployment prevalence estimate; harvested population is not a random web sample
10	The residual R4 signal is phishing-specific information	No sufficient evidence	NOT ESTABLISHED	Unresolved	Above-chance transfer is consistent with, but does not demonstrate, phishing-specific information
11	A PhiUSIIL construction mechanism causing the effect is identified	No experiment manipulates a candidate mechanism	NOT ESTABLISHED	Unresolved	Legitimate-class provenance is not documented at a verifiable level
12	A benchmark score can be interpreted as a universal deployment metric	Contradicted by §13.5, §13.6 and §13.11	REJECTED	Methodological	Representation, source composition and split each move the score materially

25. Appendix: Frozen Result References

Result area	Frozen file
P3 factorial AUC, decomposition, bootstrap	results/P3_source_factorial.json
Cross-source contrasts with CIs	results/P3_tables/T5b_cross_source_differences.csv
Repeated-split stability	results/P3_split_variance/P3_split_variance.json
Platform masking, 10 seeds	results/P3_platform_masking_seeds/platform_masking_seeds.json
U3 platform-grouped evaluation	results/P3_U3_platform_grouped/u3_platform_grouped.json
Cross-source transfer, shared benign	results/P3_cross_source_transfer/cross_source_transfer.json
Cross-source transfer, independent benign	results/P3_cross_source_independent_benign/independent_benign.json
Negative controls	results/P3_negative_control/negative_control_results.json

Result area	Frozen file
Cross-model families	results/P2_model_families.json
Unit definitions and concentration	results/P1b_R_unit_comparison.json
Crawler pools and funnels	results/P3_crawler_source.json
Real-world stress test	results/T8_headline_full.json
Structural audit, frozen protocol	results/manifests/P3_factorial_FROZEN_manifest_v3.json
PhishTank temporal profile	results/P3_phishtank_temporal_profile.json

26. Appendix: Provenance and Corrections

Summarised from `results/publication/audit/final_provenance_audit.md`, which is authoritative for this appendix.

Verification status.

- 115 / 115 locked values re-derived from their authoritative frozen files and matched.
- 10 / 10 frozen protected outputs hash-identical to the hashes recorded by the final experimental run.
- 18 / 18 publication artifacts hash-verified against the artifact manifest.

Benchmark false-positive rate — resolved. Two frozen files disagreed on a quantity named `D_fpr_homepages`. The value **0.00025977881689305 (7 / 26,946)** from `T8_headline_full.json` is authoritative: it is measured on the held-out, domain-disjoint test split of the full 235,795-URL corpus with the model trained in the same script, and it is independently corroborated by `B_raw_fpr` computed through a different code path in the same file. The competing value of 5e-05 (1 / 20,000) from `T7_deployment.json` is **obsolete**: it used a pre-pickled model trained on a deterministic 40,000-row subsample and evaluated it without reapplying the split, so **16,071 of its 20,000 evaluation URLs (80.36%) were in that model's training set**. T7 is not used in the scientific narrative of this report.

Statistical weighting — corrected. The methodology section previously stated only that ROC-AUC was "reported unweighted", which was ambiguous. §12.1 now states the prediction unit, grouping unit, bootstrap unit and metric weighting explicitly, together with the URLs-per-domain disparity.

Temporal sensitivity — established as final-pipeline-equivalent. The three ≤ 2023 cells predate the case-normalisation correction to the U1 grouping key but contain zero case-colliding U1 values, so their group sets, partitions, models and metrics are identical under both conventions.

Cross-model R3 naming. The frozen `P2_model_families.json` contains a `www_stripped` field (www removed, path retained) distinct from `www_and_path_stripped`, which is R3. This report uses `www_and_path_stripped` throughout and does not use the `www_stripped` value.

Structural audit. Only the three quantities recorded in the frozen manifest are reported: 200 reviewed, 0 visibly suspicious, Wilson 95% upper bound 0.019. An additional figure circulating in earlier drafts was not traceable to any frozen output and is not reported.

Modification record. No experiment was rerun. No frozen result file was modified. No experimental script was modified. Only publication builder scripts were changed, and only to correct reporting.

Outstanding repository items, none affecting any scientific result: `README.md` still describes a superseded research direction; the superseded 40,000-sample phase artifacts remain in `results/` and `models/` unmarked; the Figure 9 and Table 1 label for the benchmark FPR could be sharpened to name the held-out

test split explicitly.